

Mohammed Emad Negm

Junior Front-End Developer | React • TypeScript • 3D Web
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[GitHub](#) • [LinkedIn](#) • [Portfolio](#)



PROFESSIONAL SUMMARY

Junior front-end developer and Software Engineering student with hands-on experience building responsive web applications, interactive 3D product experiences, multi-agent AI systems, and full-stack business platforms. Skilled in React, TypeScript, JavaScript, WebGL animation, REST APIs, state management, and testing. Builds mobile-first interfaces with attention to performance, usability, and maintainable code.

TECHNICAL SKILLS

Front-End: HTML5, CSS3, JavaScript (ES6+), TypeScript, React, Responsive Web Design

3D & Animation: React Three Fiber, Drei, GSAP, ScrollTrigger, Framer Motion, Lenis

Back-End & Data: Node.js, Express, REST APIs, SQLite, MongoDB, Firebase

AI & Real-Time: LangGraph, Multi-Agent Orchestration, AI Model Routing, WebSockets

Tools & Other: Vite, Tailwind CSS, Zustand, Git, GitHub, Vitest, Docker, Flutter, Python, C++

SELECTED PROJECTS

VÉRA Fragrances - Luxury 3D E-Commerce Experience [GitHub](#)

- Built a mobile-first perfume storefront with a scroll-driven 3D bottle film, a 13-product collection, interactive product pages, cart, cash-on-delivery checkout, and an owner dashboard.
- Optimized rendering to no more than three WebGL contexts with device-tier quality scaling and API fallback. Load-tested the backend at 50 concurrent users with 2,650 requests/second, 0 errors, and p99 around 46 ms.
- **Tech:** React, TypeScript, React Three Fiber, GSAP, Framer Motion, Node.js, Express, SQLite, Vitest, Docker

SAMS - Multi-Agent Virtual AI Office

- Orchestrated Claude for implementation, GPT for planning, and Gemini for testing. Agents discuss requirements, divide projects by expertise, exchange context, review outputs, and resolve blockers as one delivery team.
- Designed an interactive virtual company with dedicated workstations, meeting rooms, a shared whiteboard, agent movement, coffee breaks, and real-time collaboration so each AI behaves like a digital employee.
- **Tech:** React, Three.js, FastAPI, LangGraph, WebSockets, Multi-Model AI Routing

Acoustic Room Mapper - Real-Time Acoustic Geometry Detection [GitHub](#)

- Built a cross-platform system that analyzes acoustic responses and generates live 2D room floor plans with sub-50 ms processing latency.
- Connected a Flutter Material 3 interface to a Python Flask REST API and C++17 DSP engine using FFT and cross-correlation. Added four visualization modes, Arabic RTL support, live metrics, and JSON export.
- **Tech:** Flutter, Dart, Python, Flask, C++17, CMake, FFT, REST APIs

Nexora ERP - Multi-Tenant Business Management Platform [GitHub](#)

- Built an ERP and SaaS platform covering sales, purchases, inventory, manufacturing, HR, CRM, logistics, treasury, finance reports, billing, and AI-assisted actions.
- Implemented tenant isolation, role-based permissions, JWT authentication, offline-first storage, cloud sync, printable invoices, analytics, and Android packaging through Capacitor.
- **Tech:** React, Vite, Node.js, Express, MongoDB, Firebase, IndexedDB, JWT, Docker, Capacitor

CERTIFICATIONS

Meta - Coursera: Introduction to Front-End Development • Programming with JavaScript • Version Control

EDUCATION

B.Sc. in Computers & Information - Software Engineering

Al-Ryada University for Science and Technology (RST) • Third Year in progress • CGPA: 3.39 / 4.0

Languages: Arabic - Native • English - B1 Intermediate